

ENHANCE YOUR BASH SCRIPTS

USING GUM AND TPUT FOR A NEXT-LEVEL EXPERIENCE



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Introduction: Why Bash Scripting Needs a Boost

Bash scripting is a cornerstone of Linux. It automates repetitive tasks, manages systems, and simplifies workflows. However, traditional scripts often feel plain and lack user interaction or visual appeal.

Imagine your script:

- Asking users for input with a stylish prompt
- Displaying colorful outputs or progress bars
- Creating menus to choose from options

This is where **Gum** and **tput** step in. These tools bring interactivity and style to Bash, making your scripts professional and user-friendly.

What is Gum?

Overview

Gum is a modern tool designed to add interactive and visually appealing elements to your scripts. With simple commands, Gum can:

- Display interactive prompts
- Add checklists or selections
- Show progress bars
- Handle inline inputs

Installing Gum

Choose the method that fits your system:

```
# macOS (Homebrew)
```

```
brew install gum
```

```
# Ubuntu/Debian (via third-party repo)
```

```
sudo apt install gum
```

```
# Arch Linux
```

```
sudo pacman -S gum
```

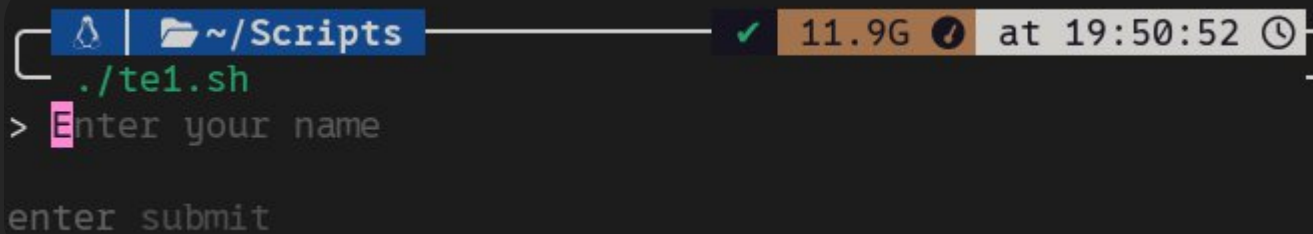
Examples of Gum in Action

1. Input Prompt

```
#!/bin/bash
```

```
name=$(gum input --placeholder "Enter your name")
```

```
echo "Hi, $name! Welcome!"
```

A terminal window with a dark background. The title bar shows a file icon, a folder icon, and the path ~/Scripts. On the right, there's a green checkmark, 11.9G, a battery icon, and the time at 19:50:52. The terminal content shows the command ./te1.sh being executed. Below it, a pink placeholder text "Enter your name" is displayed. At the bottom, the text "enter submit" is visible.

```
~/Scripts  
./te1.sh  
> Enter your name  
enter submit
```

2. Select Menu

```
#!/bin/bash
```

```
choice=$(gum choose "Option 1" "Option 2" "Option 3")
```

```
echo "You selected: $choice"
```

A terminal window with a dark background. The title bar shows a file icon, a folder icon, and the path ~/Scripts. On the right, there's a green checkmark, 11.9G, a battery icon, and the time at 19:57:48. The terminal content shows the command ./te1.sh being executed. Below it, the text "Choose:" is displayed. Then, a list of options is shown: "Option 1", "Option 2", and "Option 3". At the bottom, the text "←↑→ navigate • enter submit" is visible.

```
~/Scripts  
./te1.sh  
Choose:  
> Option 1  
Option 2  
Option 3  
←↑→ navigate • enter submit
```

3. Checkbox Menu

```
#!/bin/bash

choices=$(gum choose --no-limit "Task 1" "Task 2"
"Task 3")

echo "You chose: $choices"
```



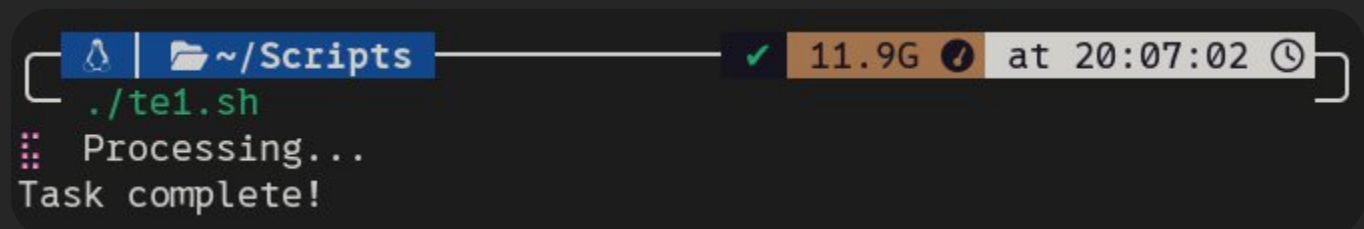
The screenshot shows a terminal window with a title bar indicating the current directory is ~/Scripts. The command ./te1.sh has been executed. The output shows a 'Choose:' prompt followed by a list of three tasks: 'Task 1', 'Task 2', and 'Task 3'. 'Task 1' and 'Task 2' are marked with checkmarks, indicating they are selected. 'Task 3' is marked with a dot, indicating it is not selected. At the bottom of the terminal, a help message for gum is visible: 'x toggle · ←↑↓→ navigate · enter submit · ctrl+a select all'.

4. Progress Bar

```
#!/bin/bash

gum spin --spinner dot --title "Processing..." --
sleep 3

echo "Task complete!"
```



The screenshot shows a terminal window with a title bar indicating the current directory is ~/Scripts. The command ./te1.sh has been executed. The output shows a 'Processing...' message with a progress bar represented by a series of dots. After a 3-second delay, the message 'Task complete!' is displayed.

5. Inline Input

```
#!/bin/bash
```

```
echo "Enter a command: $(gum input)"
```

What is **tput**?

Overview

tput is a command-line utility for styling terminal output. It lets you:

- Change text colors
- Format text (bold, underlined)
- Clear the screen or parts of it
- Control cursor placement

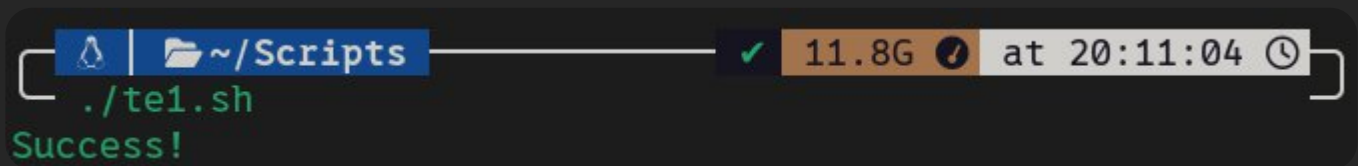
Pre-installed on Most Systems

No installation needed! **tput** is built into most Linux distributions.

Examples of tput in Action

1. Adding Color

```
#!/bin/bash
tput setaf 2  # Green text
echo "Success!"
tput sgr0     # Reset text formatting
```



2. Bold and Underlined Text

```
#!/bin/bash
echo "Normal text"
tput bold
echo "Bold text"
tput sgr0

tput smul
echo "Underlined text"
tput rmul
```

```
🔍 | 📁 ~/Scripts | ✓ 11.9G 🕒 at 20:22:57 🕒  
./te1.sh  
Normal text  
Bold text  
Underlined text
```

3. Moving the Cursor

```
#!/bin/bash  
tput cup 5 10  
echo "This text starts at row 5, column 10"
```

```
🔍 | 📁 ~/Scripts | ✓ 11.8G 🕒 at 20:31:53 🕒  
./te1.sh  
  
This text starts at row 5, column 10
```

4. Clearing the Screen or Line

```
#!/bin/bash  
echo "This will be cleared!"  
sleep 2  
tput clear
```

Combining Gum and tput for Maximum Impact

Interactive Menu with Colorful Output

```
#!/bin/bash

choice=$(gum choose "Red" "Green" "Blue")

case $choice in
    Red)
        tput setaf 1
        echo "You chose Red!"
        ;;
    Green)
        tput setaf 2
        echo "You chose Green!"
        ;;
    Blue)
        tput setaf 4
        echo "You chose Blue!"
        ;;
esac

tput sgr0
```

 |  ~/Scripts —  11.8G  at 20:41:27 

`./te1.sh`
You chose Green!

 |  ~/Scripts —  11.8G  at 20:42:25 

`./te1.sh`
You chose Red!

 |  ~/Scripts —  11.8G  at 20:42:28 

`./te1.sh`
You chose Blue!

 |  ~/Scripts —  11.8G  at 20:42:34 

 |  ~/Scripts —  11.8G  at 20:42:34 

 |  ~/Scripts —  11.8G  at 20:42:34 

`./te1.sh`

Choose:

> Red
Green
Blue

←↓↑→ navigate • enter submit

Real-World Use Cases

1. Setup Wizards

Combine Gum for input prompts and tput for formatting output to create interactive setup scripts.

2. Monitoring Tools

Use tput for colorful dashboards and Gum for interactive alerts.

3. Deployment Scripts

Display stylish progress bars and interactive menus to streamline deployments.